

Patricio Alejandro Tula

Senior Software Engineer

Brest, France
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Senior Software Engineer specialized in distributed systems and high-performance infrastructure. Experienced in designing and operating real-time, low-latency services at scale (up to Tbps traffic), with a focus on reliability, observability, and production excellence. Strong contributor to technical leadership, team maturity, and cross-functional execution in fast-growth environments.

Work Experience

Jul 2025 - Present **Senior Software Engineer – Baremetal System Network Team, OVHcloud**, Brest, Brittany, France

Modernizing and scaling legacy network management systems to improve reliability, maintainability, and operational performance in production environments. Working with Perl, Go, Kubernetes, Docker, and KEDA to build scalable production services, modernize legacy applications, and support day-to-day operations alongside SRE teams.

Key Contributions:

- Built and optimized batch processes in Perl and Go to automate configuration and management of network overlays.
- Maintained and enhanced critical legacy Perl applications, improving operational stability and scalability.
- Improved system stability by reducing failure scenarios during batch execution through better validation and error handling strategies.
- Implemented observability metrics and performance instrumentation that exposed service bottlenecks, enabling targeted optimizations and measurable performance improvements.
- Designed Kubernetes architectures, deployed them to production, and provided operational support for running services.
- Containerized infrastructure with Docker and Kubernetes, and contributed documentation, bug fixing, and troubleshooting with SRE teams.
- Supported team execution by sharing engineering practices, reviewing designs/code, and helping onboard engineers on legacy-to-modern migration topics.

Feb 2024 - Sep 2025 **Software Engineer – Network Defense Orchestration Team, OVHcloud**, Brest, Brittany, France

Built and optimized gRPC-based, event-driven applications to orchestrate OVHcloud's anti-DDoS infrastructure. Designed high-performance real-time systems in Go, MongoDB, Kubernetes, Docker, and Helm to analyze, filter, and mitigate large-scale DDoS attacks.

Key Projects & Achievements:

- Designed and deployed a low-latency gRPC streaming orchestration service handling real-time traffic analysis, reducing mitigation response time under high-load conditions.
- Designed distributed data pipelines for real-time ingestion, analysis, and filtering of network traffic, ensuring low-latency processing and consistency across services.
- Deployed and operated Kubernetes-based microservices that improved resilience under sustained attack traffic.
- Developed a high-performance rule-processing engine for real-time threat mitigation across edge, backbone, and datacenter infrastructure up to 1.3 Tbit/s.
- Worked with MongoDB and PostgreSQL to design data models supporting high-throughput ingestion and querying.
- Contributed to cross-team technical decisions and incident-readiness practices for systems protecting critical customer workloads.

- Sep 2021 - **Software Engineer – Network Automation**, OVHcloud, Paris, *Île-de-France*, France
- Feb 2024 Designed and developed APIs to manage the overlay layer of OVHcloud’s network using Go, TemporalIO, PostgreSQL, Kubernetes, Docker, and Domain-Driven Design. Maintained and optimized legacy Python applications handling the underlay network, ensuring stability and scalability.
- Key Contributions:**
- Built APIs and automation workflows to manage network overlays and improve service scalability.
 - Maintained and optimized high-traffic Python applications supporting the network underlay.
 - Automated Arista EOS 4.23 and Juniper Junos configurations with Python for repeatable network operations.
 - Advocated Agile practices and DDD adoption, improving collaboration and delivery efficiency.
 - Contributed to *Bring Your Own IP (BYOIP)* and *Additional IP*, enabling flexible IPv4 and IPv6 allocation and migration workflows.
 - Coordinated delivery between product and engineering stakeholders to structure execution on multi-service roadmap items.
- Jul 2019 - **DevOps Engineer**, ORNESS, Paris, France
- Sep 2021 Ran and monitored performance tests on new and existing software with Prometheus and Grafana to isolate bottlenecks, debug issues, and identify improvement areas. Developed infrastructure automation with Terraform and Kubernetes, maintained backend applications in Python/Flask, enhanced Jenkins CI/CD pipelines in Groovy, and contributed to OpenStack source code in a strong Scrum environment.
- Jan 2018 - **Full Stack Engineer**, Ekumen (*InOrbit*), Buenos Aires, Argentina
- Dec 2018 Designed, developed, tested, and debugged a robotic fleet control cloud application using ROS. Built frontend and backend features with ReactJS, Node.js, and Jest; improved CI/CD with Jenkins, Docker, and AWS ECS; and developed simulations and testbenches for robotic navigation with Python.
- Jul 2017 - **R&D Software Intern**, IMT Atlantique, Brest, France
- Jan 2018 Completed a six-month internship funded by the European program “Santé, autonomie et bien-être”. Researched and developed a production-like cloud application for Brest’s medical center (CHRU Brest) and published the paper “*An implementation of an imitation game with ASD children to learn nursery rhymes.*” Technologies used: JavaScript, Python, HTML, CSS, NodeJS, gRPC, and C++.
- Jul 2015 - **Junior Full Stack Developer**, Ideamatic, Buenos Aires, Argentina
- Jan 2017 Designed and implemented a 3D graphics rendering application in JavaScript, with ReactJS on the frontend and PHP/CakePHP on the backend.
- Sep 2014 - **Teaching Assistant**, Universidad de Buenos Aires, Buenos Aires, Argentina
- Aug 2016 Taught practical algorithmics lessons for second-year electronic engineering students in C++, covering sorting, merging, and time/space complexity.

Education

- 2019 **Telecom Engineer**, IMT-Atlantique (*Telecom Bretagne*), Brest, France
Focused on telecommunications, networking, and software systems.
- 2017 **Electronic Engineer**, Universidad de Buenos Aires, Buenos Aires, Argentina
Worked on projects emphasizing networking, distributed systems, and embedded development.
- 2007 **Baccalauréat Scientifique**, Lycée Franco-Argentin Jean Mermoz, Buenos Aires, Argentina
Graduated with a focus on scientific and technical studies in a Franco-Argentinian curriculum.

Programming and Computer Skills

	Level	Skill	Years	Comment
Languages:	■■■■■	Go (Golang)	8	Backend services, network automation, gRPC, concurrency, distributed systems
	■■■■■	Python 	7	Automation, Flask services, networking, performance tooling, OpenStack
	■■■■■	Perl	6	Batch processing, legacy system maintenance, network overlay management
	■■■■■	JavaScript 	5	Node.js, ReactJS, Jest
	■■■■■	C, C++	4	System-level programming, robotics (ROS), microcontrollers
OS:	■■■■■	Linux 	8	Scripting (bash/zsh), observability, production operations

Distributed Systems	Microservices, event-driven architecture, asynchronous workflows, TemporalIO	Protocols & APIs	gRPC, REST, OpenAPI
DevOps	Docker 🐳, Kubernetes, Helm, Terraform, Jenkins, KEDA	Databases	PostgreSQL, MongoDB
Architecture	Domain-Driven Design, testing practices, reliability	Monitoring	Prometheus, Grafana, Graylog, Loki, Tempo
Cloud	AWS, OVHcloud, OpenStack	Leadership	Mentoring, code reviews, delivery structuring, cross-team collaboration
Operational Focus	Performance, resilience, scalability, automation	VCS	Git git , GitHub/GitLab

Systems & Performance

- Core Competencies
- Distributed systems design (event-driven, streaming)
 - High-throughput / low-latency systems
 - Observability & performance tuning
 - Fault tolerance & resilience under load

Languages **A** **z**

- Spanish Native
- French Bilingual
- English B2

Publications

- 2018 “**An implementation of an imitation game with ASD children to learn nursery rhymes.**”, *Internship Research at IMT Atlantique*
 Paper published as part of a European-funded research project in Brest, France.

Personal Projects

- 2015 **LeapTower**, *GameOn*, Buenos Aires, Argentina
LeapTower is an experimental video game built alongside Federico Marino for the tech event “Game On - El arte de jugar.” Exhibited at the Centro Cultural San Martín for a week, with over 1,000 visitors playing it.
 Blog Post: <http://www.fedemarino.com.ar/leaptower-2/>